

Mezzan / STC — Integration Reference Handbook

Zoho Inventory · Organisation 150001417597 · prepared by Hallek ERP Consulting
For: Mohamed Waqqas, Integration Development

How to use this document

This is the single reference for the integration build. It covers the warehouse structure the transactions operate within, the inbound and outbound cycles step by step, every custom field and why it exists, the STC API files and what each maps to, and the decisions still outstanding.

It supersedes nothing — it consolidates. The field inventory workbook issued previously remains the searchable reference for api_names; this document explains what those fields are for and how the transactions behave.

Read Section 5 before writing any mapping. Three fields carry identical on-screen labels but different api_name values depending on the module. Zoho accepts a payload containing a wrongly-named custom field, returns HTTP success, and silently discards the value. These failures do not surface in testing unless results are verified against the organisation.

1. Warehouse structure

Every transaction resolves to a location, a zone and a bin. Understanding this first makes the rest of the document straightforward.

1.1 Location hierarchy

There is a parent location, STC warehouses, which holds the physical address and has bin tracking disabled. It is a container, not a stock-holding location.

Beneath it sit six child locations, one per STC entity. Each has its own vendor and customer record — twelve contacts in total.

Entity	Location ID	Bins built
Solutions by STC	49066000000116096	Full — ~3,488 bins
Upsource by Solution	49066000000121100	Zones only
Pulse by Solution	49066000000121122	Zones only
VAS by Solution	49066000000121142	Zones only
Solution Operation Project	49066000000121162	Zones only
STC Disposal Project	49066000000121182	Zones only

1.2 Zones

Each entity carries eighteen zones: three cities (Riyadh, Dammam, Jeddah) × two pools (standard, FreeStock) × three sub-inventories (Project, Internal, Unused).

The zone name is exactly the value the Sub Inventory and Intended Sub-Inventory dropdowns accept. A value outside that list is rejected.

Each entity also carries three Zoho default zones created at setup, including a legacy “Receiving Zone” holding a bin called “Receiving Area”. That is not part of this design — never target it.

1.3 Bins

Every zone has one RECEIVING bin, named “{City} - {Sub-Inventory} - RECEIVING”. Physical shelf bins follow the same pattern with the locator appended, for example “Riyadh - Project - MZNAC0100A”. The spaces either side of each hyphen are part of the name.

The integration writes stock ONLY to the eighteen RECEIVING bins. It never targets a physical shelf location. Movement onto shelves is a Move Order, performed manually by Mezzan, and is outside Phase 1 scope. Move Order is not a workflow-capable entity in Zoho and cannot carry custom fields.

1.4 Resolving a sub-inventory to a bin

City is Dammam when the locator's fourth segment is Dammam, otherwise Riyadh. Sub-inventory maps Project to Project, InternalN to Internal, Internal to Internal, UnUsed to Unused. Target bin is “{City} - {SubInv} - RECEIVING”.

2. Inbound cycle

Four steps, two of which are integration touchpoints. There is no separate ASN object in Zoho — an ASN becomes a Purchase Receive.

Step	STC endpoint	Zoho object	Stock effect
1	AddPurchaseOrder	Purchase Order	None
2	AddASN	Purchase Receive (Pending Approval)	None — nothing posts
3	— (Mezzan approve in Zoho)	Purchase Receive (Received)	Stock posts to RECEIVING bin; batch created
4	AddGRN (outbound)	— fires on approval	Already posted at step 3

2.1 Why the approval gate exists

The ASN tells Mezzan goods are coming. The approval is Mezzan confirming they physically arrived. Stock must not post before that, or the system shows inventory that is not on the floor. Do not auto-approve or design around this.

2.2 What happens on approval

- Stock posts into the RECEIVING bin of the zone named in Sub-Inventory
- A batch is created, with the reference constructed as described in Section 6
- AddGRN fires back to STC

PO Number on the receipt is native, inherited from the purchase order and locked. Receipt number is native and auto-generated on its own series.

2.3 Put-away — for context, not for build

Stock then sits in the receiving bin until Mezzan move it onto physical shelves by Move Order import. This is deliberate: STC locator values have no bridge to Mezzan's MZNxxxx barcodes, so nothing in STC's data can determine which shelf a carton belongs on. Only the person placing it knows.

3. Outbound cycle

Step	STC endpoint	Zoho object	Stock effect
1	AddSalesOrder / CreateRFD	Sales Order	Committed, not dispatched
2	— (Mezzan pick and pack)	Package	Batch selected at picking
3	— (Mezzan ship)	Shipment Order	Stock leaves
4	SendShippingConfirmation	— fires on dispatch	Already deducted at step 3

3.1 Batch selection at picking

Every item is batch tracked, so the picker selects which batch is being dispatched. This has an operational consequence worth knowing: because each distinct combination of received date, locator and supplier item code becomes its own batch, a single item can present a long list at this point. One item in the current data carries seventy separate batches.

Serial tracking is OFF across the organisation. TrackSerialNumber is false on all items. Serial numbers are not part of any flow — do not map them.

3.2 Reporting back

cf_batch_id on the Shipment Order carries the dispatched lot back to STC. Since everything is batch tracked, it should always be populated.

4. Custom fields — what they are for

The field inventory workbook lists every `api_name`. This section explains why they exist, which is what determines correct mapping.

4.1 The four STC reference fields

STC asked that PO No, Line No, Project No and Release No be captured so they can track shipment, dispatch and reporting. These are held at LINE level, not header, because a single receipt can cover several PO lines with different values.

Field	api_name	Why line level
PO Line Number	cf_vendor_po_line_number	Differs per line by definition
Project Number	cf_project_number	One receipt may span projects
Release Number	cf_release_number	Follows the PO line
Supplier's Item Code	cf_supplier_s_item_code	Item-specific. MANDATORY.

cf_supplier_s_item_code is MANDATORY at line level on item, purchaseorder_item, purchase_receive_item and salesorder_item. Any payload creating those records without it will be rejected.

4.2 Locator

`cf_storage_location` at Purchase Receive line level holds STC's locator. It exists at line level because STC locators vary per line — the second segment is a line sequence. The header fields `cf_locator_full` and `cf_locator_location` cannot represent a multi-line receipt.

The locator is a five-segment string: Project . Line Number . PO Number . Location . Shipset Number. For example 16602.31.41294.DB Schenker.00000

4.3 Restricted dropdowns

Four fields accept only listed values. Anything else is rejected outright.

Field	Entity	Allowed values
Sub Inventory	item, PO line, PR line, SO line	The 18 zone names
Intended Sub-Inventory	purchaseorder (header)	The 18 zone names
Transaction Priority	salesorder (header)	High / Medium / Low
Ship With Care	salesorder (header)	Yes / No

5. API name traps — read before mapping

Zoho accepts a payload containing a wrongly-named custom field, returns success, and discards the value. These do not surface in testing unless results are verified against the organisation rather than the HTTP response.

5.1 PO Line Number has two api_names

Entity	api_name
item	cf_vendor_po_line_number
purchaseorder_item	cf_vendor_po_line_number
purchase_receive_item	cf_vendor_po_line_number
salesorder_item	cf_vendor_po_line_number
salesorder (HEADER)	cf_po_line_number
shipment_order	cf_po_line_number

5.2 Locator has three api_names

Entity	api_name	Label
purchase_receive_item	cf_storage_location	Locator — USE THIS
purchase_receive (header)	cf_locator_full	Locator (full)
purchase_receive (header)	cf_locator_location	Locator Location

5.3 Sales Order carries two PO number fields

cf_po_number and cf_locator_po_number both exist on the Sales Order header. Which one the integration populates is an open decision — see Section 8.

5.4 Discovering fields correctly

Header and line fields live under different entity names. Querying list_custom_fields on the parent returns item_entity_fields empty, which is misleading — line fields live under the separate *_item entity. list_custom_fields_simple returns empty for line entities and cannot be relied on.

6. Batch reference construction

Agreed with STC so they can see stock categorised by their own identifying information.

{Date Received} - {Locator} - {Supplier Item Code}

Example: 03/03/2026 - 16602.31.41294.DB Schenker.00000 - 6536501-1

- Separator is a SPACED hyphen. STC supplier codes contain hyphens themselves, so an unspaced separator makes the parts ambiguous.
- The locator goes in EXACTLY as STC supply it — full five segments, never shortened or parsed.
- Build from the LINE-level locator, not the header.
- Date is dd/MM/yyyy. The native manufactured_date field carries the same received date as yyyy-MM-dd.

Consequence: every distinct date/locator/supplier combination becomes its own batch. Expect high batch counts — roughly 2,000 batches across 782 items in the Solutions data, one item carrying 70.

7. STC API files — what maps where

Six endpoints in Phase 1, drawn from the DBSchenker v4.3 and ERP-SupplyChain v3.1 / v2.0 specifications.
Test endpoint `apitest.stcs.com.sa`

Endpoint	Direction	Creates in Zoho	Key mapping notes
AddItemRequest	STC → Zoho	Item	SKU must carry the STC item code — items are searchable in transactions by name or SKU only, never by custom field. Supplier's Item Code mandatory.
AddPurchaseOrder	STC → Zoho	Purchase Order	Set status Issued for open POs; default is Draft. Intended Sub-Inventory mandatory at header.
AddASN	STC → Zoho	Purchase Receive (Pending Approval)	Must carry locator PER LINE. Sub-Inventory mandatory at header. No stock posts.
AddGRN	Zoho → STC	—	Fires on Mezzan approval of the Purchase Receive. Stock posts at that moment.
AddSalesOrder / CreateRFD	STC → Zoho	Sales Order	Two restricted dropdowns — Transaction Priority, Ship With Care.
SendShippingConfirmation	Zoho → STC	—	Fires on dispatch. <code>cf_batch_id</code> carries the dispatched lot.

8. Open decisions

These must be answered before the mapping can be finalised. They are not optional.

#	Decision	Owner	Why it blocks
1	Sales Order header: populate <code>cf_po_number</code> or <code>cf_locator_po_number</code> ?	STC	Both exist. Wrong choice produces reconciliation mismatches.
2	Does the ASN payload carry the locator per line?	STC	The batch reference depends on it, and locators vary by line.
3	Do STC's Transaction Priority and Ship With Care values match the dropdown lists exactly?	STC	A value outside the list is rejected outright.

9. Verification checklist

Before declaring any endpoint complete:

- Line-level custom fields populate on EVERY line, not just the first — the commonest failure is a value landing on the header only
- `cf_supplier_s_item_code` present on every line of every payload
- Sub-inventory resolves to one of the eighteen exact zone names
- Stock lands in the correct RECEIVING bin, never a physical locator
- Batch reference matches the Section 6 format exactly, including spaced separators
- Batch reference built from the LINE-level locator, tested on a multi-line receipt where locators differ
- ASN creates a receipt in pending approval with no stock movement
- GRN fires only on approval, and stock posts at that point
- PO Line Number targets the correct `api_name` for the module under test
- A partial receipt leaves the PO partially_received with the correct balance open

Verify posted results against the organisation, not the HTTP response. A 200 does not mean the custom fields landed.

10. Environment

Item	Value
Organisation ID	150001417597 — required on every API call
URL	inventory.zoho.sa
Test endpoint	apitest.stcs.com.sa
Specifications	DBSchenker APIs v4.3 · ERP-SupplyChain APIs v3.1 / v2.0
Batch tracking	ON for every item
Serial tracking	OFF — do not map
Bin tracking	ON